

ME462 Loop Closure Kinematic Analysis

Little Daisies

April 2026

Kinematic Model

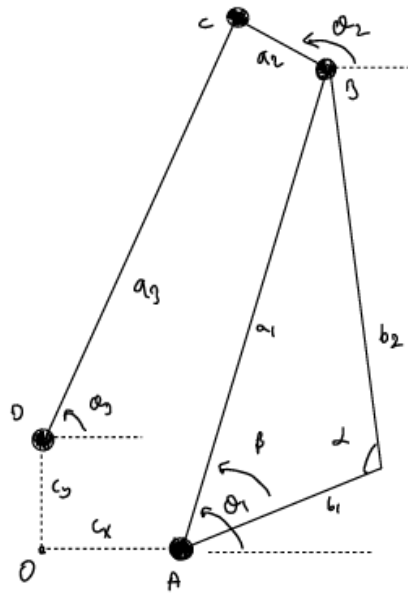


Figure 1: Kinematic system.

The loop closure equation is

$$\vec{OA} + \vec{AB} + \vec{BC} + \vec{CD} + \vec{DO} = 0$$

or in complex form,

$$c_x + a_1 e^{i\theta_1} + a_2 e^{i\theta_2} - a_3 e^{i\theta_3} - c_y i = 0$$

Separating into real and imaginary parts:

$$c_x + a_1 \cos \theta_1 + a_2 \cos \theta_2 - a_3 \cos \theta_3 = 0 \quad (1)$$

$$a_1 \sin \theta_1 + a_2 \sin \theta_2 - a_3 \sin \theta_3 - c_y = 0 \quad (2)$$

Solution for Input θ_1

Define

$$C_1 = c_x + a_1 \cos \theta_1$$

$$C_2 = a_1 \sin \theta_1 - c_y$$

Then (1) and (2) become

$$a_3 \cos \theta_3 = C_1 + a_2 \cos \theta_2$$

$$a_3 \sin \theta_3 = C_2 + a_2 \sin \theta_2$$

Squaring and adding,

$$a_3^2 = (C_1 + a_2 \cos \theta_2)^2 + (C_2 + a_2 \sin \theta_2)^2$$

$$a_3^2 = C_1^2 + C_2^2 + a_2^2 + 2a_2(C_1 \cos \theta_2 + C_2 \sin \theta_2)$$

Therefore,

$$C_1 \cos \theta_2 + C_2 \sin \theta_2 = \frac{a_3^2 - a_2^2 - C_1^2 - C_2^2}{2a_2}$$

Define

$$d = \sqrt{C_1^2 + C_2^2}$$

$$\phi = \text{atan2}(C_2, C_1)$$

Since

$$C_1 = d \cos \phi, \quad C_2 = d \sin \phi$$

we obtain

$$d \cos(\theta_2 - \phi) = \frac{a_3^2 - a_2^2 - d^2}{2a_2}$$

Hence,

$$\theta_2 = \phi \pm \cos^{-1} \left(\frac{a_3^2 - a_2^2 - d^2}{2a_2 d} \right)$$

After finding θ_2 , θ_3 is found from

$$\theta_3 = \text{atan2}(C_2 + a_2 \sin \theta_2, C_1 + a_2 \cos \theta_2)$$

Solution for Input θ_3

For a given input θ_3 , define

$$K_1 = a_3 \cos \theta_3 - c_x$$

$$K_2 = c_y + a_3 \sin \theta_3$$

Then (1) and (2) become

$$a_1 \cos \theta_1 + a_2 \cos \theta_2 = K_1$$

$$a_1 \sin \theta_1 + a_2 \sin \theta_2 = K_2$$

Rearranging,

$$a_1 \cos \theta_1 = K_1 - a_2 \cos \theta_2$$

$$a_1 \sin \theta_1 = K_2 - a_2 \sin \theta_2$$

Squaring and adding,

$$a_1^2 = (K_1 - a_2 \cos \theta_2)^2 + (K_2 - a_2 \sin \theta_2)^2$$

$$a_1^2 = K_1^2 + K_2^2 + a_2^2 - 2a_2(K_1 \cos \theta_2 + K_2 \sin \theta_2)$$

Therefore,

$$K_1 \cos \theta_2 + K_2 \sin \theta_2 = \frac{K_1^2 + K_2^2 + a_2^2 - a_1^2}{2a_2}$$

Define

$$s = \sqrt{K_1^2 + K_2^2}$$

$$\psi = \text{atan2}(K_2, K_1)$$

Since

$$K_1 = s \cos \psi, \quad K_2 = s \sin \psi$$

we obtain

$$s \cos(\theta_2 - \psi) = \frac{s^2 + a_2^2 - a_1^2}{2a_2}$$

Hence,

$$\theta_2 = \psi \pm \cos^{-1} \left(\frac{s^2 + a_2^2 - a_1^2}{2a_2s} \right)$$

After finding θ_2 , θ_1 is found from

$$\theta_1 = \text{atan2}(K_2 - a_2 \sin \theta_2, K_1 - a_2 \cos \theta_2)$$

Distance Calculation

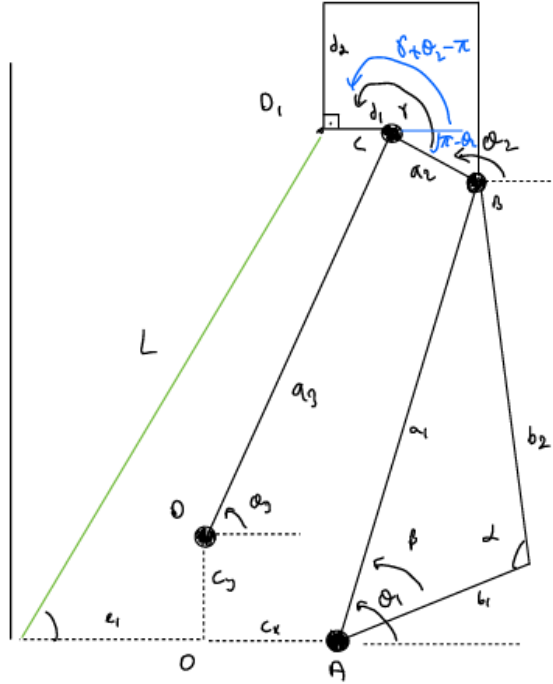


Figure 2: Distance definition for the orange part.

The position of the orange point is written as

$$e_1 + c_y i + a_3 e^{i\theta_3} + d_1 e^{i(\gamma + \theta_2 - \pi)} = L_x + L_y i$$

Separating into real and imaginary parts gives

$$L_x = e_1 + a_3 \cos \theta_3 + d_1 \cos(\gamma + \theta_2 - \pi)$$

$$L_y = c_y + a_3 \sin \theta_3 + d_1 \sin(\gamma + \theta_2 - \pi)$$

Therefore, the distance is

$$L = \sqrt{L_x^2 + L_y^2}$$

Final Form

For input θ_1 :

$$C_1 = c_x + a_1 \cos \theta_1$$

$$C_2 = a_1 \sin \theta_1 - c_y$$

$$d = \sqrt{C_1^2 + C_2^2}$$

$$\phi = \text{atan2}(C_2, C_1)$$

$$\theta_2 = \phi \pm \cos^{-1} \left(\frac{a_3^2 - a_2^2 - d^2}{2a_2d} \right)$$

$$\theta_3 = \text{atan2}(C_2 + a_2 \sin \theta_2, C_1 + a_2 \cos \theta_2)$$

For input θ_3 :

$$K_1 = a_3 \cos \theta_3 - c_x$$

$$K_2 = c_y + a_3 \sin \theta_3$$

$$s = \sqrt{K_1^2 + K_2^2}$$

$$\psi = \text{atan2}(K_2, K_1)$$

$$\theta_2 = \psi \pm \cos^{-1} \left(\frac{s^2 + a_2^2 - a_1^2}{2a_2s} \right)$$

$$\theta_1 = \text{atan2}(K_2 - a_2 \sin \theta_2, K_1 - a_2 \cos \theta_2)$$

The feasibility conditions are

$$|a_3 - a_2| \leq d \leq a_3 + a_2$$

for input θ_1 , and

$$|a_1 - a_2| \leq s \leq a_1 + a_2$$

for input θ_3 .